

Practical - 4.

Name :- Shreya Shedi

PRN :- B25ET1193.

Batch :- C3

Q1. Define data, data item and entity.

→ Data are simple values or sets of values. Data item is single unit of values. Entity is something which has certain attributes which may be assigned values.

Attribute	Name	Age	City
Value	xyz	16	Pune

Q2. Define data structure.

→ A data structure is a way of organizing and storing data in a computer so that it can be accessed and used efficiently.

Q3. What are the types of data structure? Give examples.

→ Data structures are mainly primitive and non-primitive. Non-primitive structures are further divided into linear (array, linked list) and non-linear (tree, graph).

Q4. What is nested structure?

→ A nested structure is a structure that contains another structure as one of its members. In simple words, it is a structure within a structure.

Q5. What are unions?

→ A union is a user-defined data type in C that allows different types of data to be stored under one name. All members share the same memory location.

Q6. Compare structure with unions.

→ In a structure, each member has separate memory and all members can store values at the same time. In a union, all members share the same memory so only one member can hold a meaningful value at a time.

Q7. What are operators used to access data from structures?

→ The dot (.) operator is used to access members through a structure pointer.

Q8. What is array of structure?

→ An array of structures is a collection of multiple structure variables of the same type. It is used in database management, records and arrays of structures.

Q9. List out applications of structures.

→ Structures are used to store related data of different types together. They are commonly used in database management, records and arrays of structure.

Q10. How many to access structure member through pointer?

→ A structure member can be accessed through a pointer using the arrow (→) operator. For example, ptr → member is used to access a member through a structure pointer.